AN AR-BASED COACHING SYSTEM FOR KABADDI USING POSE ESTIMATION AND LLM FEEDBACK

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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Abstract

Many Kabaddi players, particularly beginners, often practice without direct supervision from a coach. In such settings, it becomes difficult to judge whether body posture, balance, and movement timing are being executed correctly. Minor errors in joint alignment or movement coordination may go unnoticed during self-practice and can negatively affect skill development over time.

To support learners in these situations, this project proposes an AR-based Kabaddi Ghost Trainer that visually demonstrates correct techniques and provides automated feedback. The system begins by extracting human body poses from recorded videos of expert Kabaddi players using deep learning–based pose estimation methods. These raw pose outputs are then cleaned, normalized, and organized to form a reference movement sequence, referred to as the "ghost" motion.

During training sessions, the ghost motion is displayed within an augmented reality environment, allowing learners to observe and follow the expert movement while practicing. User-performed movements are recorded and analyzed offline by comparing joint alignment, posture consistency, and temporal execution against the reference sequence. Instead of relying solely on numeric similarity scores, the system interprets structured pose error metrics to generate understandable feedback through an LLM-based module. This design choice enables flexible and contextual feedback generation without embedding heuristic rules directly into the evaluation pipeline.

Overall, the project integrates pose estimation, temporal alignment, augmented reality visualization, and data-driven feedback generation to assist independent Kabaddi practice and technique improvement.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
AR	Augmented Reality
COCO	Common Objects in Context
CNN	Convolutional Neural Network
DTW	Dynamic Time Warping
EMA	Exponential Moving Average
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
MP	MediaPipe
NPY	NumPy Array File Format
TTS	Text-to-Speech
YOLO	You Only Look Once

Chapter 1

INTRODUCTION

1.1 OVERVIEW

Kabaddi is a physically demanding sport that requires precise coordination between body posture, timing, balance, and spatial awareness. Effective execution of techniques such as raids, touches, and evasive movements depends not only on physical strength but also on accurate replication of expert movement patterns. Traditional training methods largely rely on in-person coaching, verbal instructions, and repeated observation of demonstrations, which may not always provide objective or measurable feedback on performance quality.

In situations where direct supervision is limited, learners often struggle to identify subtle posture deviations or timing errors during practice sessions. This project addresses the need for an intelligent training support system by integrating computer vision, pose analysis, and augmented reality technologies. The core idea is to provide a structured and measurable training environment where expert movements can be visualized and compared against user-performed actions.

By focusing on pose-level analysis rather than raw video comparison, this work emphasizes explainability and consistency in performance evaluation. The system is designed to function as a virtual training aid that assists learners in understanding movement accuracy, identifying deviations, and improving technique execution in a systematic and data-driven manner.

1.2 PROBLEM STATEMENT

Learning and refining Kabaddi techniques requires continuous feedback on body posture, joint coordination, and temporal execution of movements. Existing training practices depend heavily on subjective observation by coaches or self-assessment through video playback, both of which lack quantitative evaluation. These approaches make it difficult to consistently identify precise movement errors, especially minor joint-level deviations that significantly impact technique effectiveness and balance during gameplay.

There is a clear need for a training system that can objectively analyze human movement and compare it against expert-defined standards without requiring constant human supervision. A key technical challenge lies in handling variations in execution speed and movement pacing, where the same technique may be performed faster or slower while preserving its structural order. Additionally, small joint-level deviations are often difficult to detect through video replay alone, making manual self-assessment unreliable. The absence of a pose-driven, temporally aligned feedback mechanism limits effective independent practice and slows skill acquisition, particularly for technique-focused sports such as Kabaddi.

1.3 OBJECTIVES

The primary objective of this project is to design and develop an intelligent Kabaddi training system that enables objective evaluation of player movements using pose-based analysis. Specifically, the objectives are:

To capture and process user-performed Kabaddi actions in a structured manner suitable for computational analysis.

To normalize and align human pose data in order to handle variations in execution speed, body proportions, and camera perspectives.

To quantitatively measure deviations between expert reference movements and user executions at both joint and temporal levels.

To localize movement errors across different phases of a technique, enabling precise identification of incorrect posture or timing.

To generate interpretable performance metrics that can support meaningful LLM-based feedback, while intentionally avoiding rule-based coaching heuristics to preserve flexibility and extensibility.

1.4 SOLUTION OVERVIEW

The proposed solution is an augmented reality–assisted training system that evaluates Kabaddi techniques through structured pose analysis rather than direct video comparison. The system is designed to separate visual demonstration, movement capture, and performance evaluation into well-defined stages, ensuring clarity and consistency in assessment. Expert movements are represented as reference pose sequences, which serve as standardized benchmarks for evaluating user-performed actions.

At the input level, user practice sessions are recorded using a mobile device, and the captured video is treated solely as a source for pose extraction. Raw video frames are not used directly for comparison or scoring. Instead, pose estimation outputs are converted into a normalized skeletal representation, enabling stable analysis independent of lighting conditions, background variations, or camera placement.

To address differences in execution speed and motion pacing, the solution incorporates temporal alignment techniques that synchronize user movements with expert reference sequences. This alignment ensures that fast or slow executions are evaluated fairly by comparing corresponding motion phases rather than absolute time frames. As a result, performance evaluation focuses on movement quality

instead of execution duration.

Following temporal alignment, the system performs joint-wise and frame-wise error computation to quantify deviations between user and expert poses. These errors are aggregated into structured metrics that highlight spatial inaccuracies and temporal inconsistencies across different phases of a technique. While evaluation and feedback generation are separated conceptually, both stages rely on shared error representations to ensure consistency. The final output consists of quantitative performance scores and interpretable error indicators that serve as inputs for feedback generation. This design balances modularity with practical integration, enabling explainable and extensible system behavior grounded in measurable data.

Overall, the proposed approach provides a systematic and objective framework for skill assessment and improvement in Kabaddi training.

1.5 ORGANIZATION OF THE REPORT

The report is organized into six chapters, each addressing a specific aspect of the project in a structured and logical manner, supported by appropriate system diagrams and technical explanations. The chapters are outlined as follows:

Chapter 1 introduces the context and motivation behind the proposed AR-based Kabaddi training system. It presents the problem statement, clearly defining the limitations of existing training approaches, and outlines the objectives of the project. This chapter also provides a high-level overview of the proposed solution and establishes the scope and direction of the work.

Chapter 2 presents a comprehensive literature survey of existing research related to human pose estimation, motion analysis, temporal alignment techniques, and technology-assisted sports training systems. It reviews prior methodologies, highlights key observations drawn from existing work, discusses their limitations,

and identifies the research gap that motivates the proposed approach.

Chapter 2

LITERATURE SURVEY

This chapter presents a focused review of existing research related to human movement analysis, pose estimation, and intelligent feedback systems in sports and activity training contexts. The surveyed literature highlights the progression from traditional video-based analysis methods to structured pose-driven and learning-based approaches, emphasizing objective evaluation and quantitative assessment of movement quality. By examining relevant studies, this chapter establishes the technical foundation upon which this project is built and identifies how prior work has addressed similar challenges in motion comparison and feedback generation. The insights derived from these studies help position the proposed system within the broader research landscape while clarifying the specific gaps that motivate this work.

2.1 EXISTING WORK

Human motion analysis has been extensively studied across biomechanics, activity recognition, and sports performance evaluation, with increasing emphasis on markerless and pose-based representations. Aleksić et al. [1] conducted a detailed validation of markerless pose estimation by comparing it against gold-standard 3D marker-based motion capture systems for biomechanical temporal analysis. Their work demonstrates strong agreement between the two approaches, indicating that skeletal joint trajectories derived from markerless systems are sufficiently accurate for quantitative motion analysis. However, the study is restricted to controlled jump-based tasks and relies on multi-camera setups, limiting its applicability to unconstrained, sport-specific movements performed in real-world

environments.

Pose estimation combined with machine learning techniques has also been applied to temporal activity analysis. Lin et al. [5] proposed a pipeline integrating OpenPose with recurrent neural network models for fall detection from video sequences. The study confirms that pose-derived temporal features can effectively capture human motion dynamics and achieve high classification accuracy. While this work validates the robustness of pose-based temporal modeling, it focuses on event detection rather than detailed evaluation of movement quality or comparison against expert reference executions, which are essential for technique-oriented sports training systems.

Skeleton-based action recognition has further advanced the use of structured skeletal representations. Devzel et al. [2] introduced a one-shot skeleton-based action recognition framework using graph convolutional networks for strength and conditioning exercises. Their approach demonstrates that skeletal motion patterns can be discriminative even with limited training data. Despite its effectiveness in recognizing action categories, the framework emphasizes classification outcomes and does not address continuous similarity measurement, temporal alignment, or joint-level error localization required for movement correction and coaching applications.

Real-time sports performance analysis has been explored by Long Liu et al. [6], who integrated OpenPose with DeepSORT to enable multi-person pose estimation and tracking in volleyball scenarios. The system handles occlusion and player interaction while maintaining real-time performance, supporting the feasibility of pose-based pipelines in dynamic sports environments. However, the primary focus remains on tracking and performance statistics rather than individualized comparison of player movements against expert-defined standards.

Mixed reality–based instructional systems have also leveraged pose and motion data to enhance physical training. Ihara et al. [3] proposed a framework

that converts 2D instructional videos into three-dimensional mixed reality avatars by extracting motion trajectories and rendering them in an immersive environment. This approach highlights the effectiveness of visual augmentation for learning physical skills. Nevertheless, it requires specialized hardware and does not provide automated, quantitative evaluation or personalized corrective feedback based on skeletal similarity metrics.

A critical methodological component across motion comparison studies is temporal alignment of movement sequences. Human actions are often performed at varying speeds while preserving their structural order, making direct frame-to-frame comparison unreliable. Dynamic Time Warping (DTW) has been established as a standard non-linear temporal alignment technique for such scenarios. Müller [7] formally presented DTW as a robust method for aligning motion and gesture sequences with different temporal rates, and it has since been widely adopted in motion analysis, gait studies, and biomechanics. Keogh and Ratanamahatana [4] further reinforced DTW's theoretical foundation for exact alignment of time-series data with temporal distortions.

In biomechanics and pose-based motion analysis, DTW is frequently used either explicitly or implicitly to establish temporal correspondence between joint trajectory sequences prior to comparison. Studies such as Aleksić et al. [1], which compare time-series joint data across different measurement systems, inherently require temporal normalization to ensure meaningful correspondence. DTW represents the standard non-linear solution for this requirement in motion analysis literature. Following temporal alignment, spatial similarity between poses is commonly quantified using Euclidean distance between corresponding joint positions or angles. This combination of DTW for temporal alignment and Euclidean distance for spatial comparison is a well-established and interpretable strategy in human motion analysis and action comparison tasks [7, 8]. It enables objective evaluation of movement quality by separating timing variability from structural pose deviation, thereby forming a reliable basis for pose-driven training

and assessment systems.

Despite these advances, none of the reviewed systems provide a unified framework that combines temporal alignment, joint-level error localization, and interpretable feedback specifically tailored for corrective sports training.

2.2 OBSERVATIONS FROM EXISTING WORK

The reviewed literature consistently demonstrates that pose-based representations are effective for modelling human motion across a wide range of applications, including biomechanics, activity recognition, and sports performance analysis. Markerless pose estimation systems have been shown to provide sufficiently accurate skeletal data for quantitative analysis when compared against traditional motion capture setups. This establishes skeletal joint trajectories as a reliable abstraction for movement evaluation, particularly in scenarios where intrusive sensors or multi-camera installations are impractical.

Another key observation is the widespread reliance on temporal modelling to handle variability in human motion execution. Studies involving action recognition, performance monitoring, and biomechanical comparison implicitly or explicitly account for differences in execution speed through temporal normalization or alignment techniques. Dynamic Time Warping emerges as a recurring methodological choice in classical motion analysis literature for synchronizing time-series data, enabling fair comparison of motion phases rather than absolute time indices.

2.3 LIMITATIONS OF EXISTING WORK

Despite significant progress in pose estimation and motion analysis, several limitations remain evident in existing research. Many pose-based systems are

evaluated under controlled conditions or rely on constrained datasets, which limits their robustness when applied to unconstrained, real-world sports movements. Variations in camera placement, execution style, and environmental factors often degrade performance, making it difficult to generalize these systems for independent training scenarios without careful adaptation.

Another notable limitation is the predominant focus on action recognition and event detection rather than movement quality assessment. Most existing approaches are designed to classify actions or detect anomalies, offering binary or categorical outputs. Such formulations do not support fine-grained evaluation of technique execution, nor do they provide detailed insights into joint-level deviations or phase-wise errors that are essential for corrective sports training.

2.4 SUMMARY OF LITERATURE SURVEY

The literature survey highlights a clear evolution from traditional video-based observation methods toward pose-driven and learning-based approaches for human motion analysis. Prior research confirms that markerless pose estimation can reliably capture skeletal joint trajectories suitable for quantitative evaluation, providing a practical alternative to sensor-based motion capture systems. Temporal modeling and alignment have been identified as essential components for handling natural variability in human movement, with Dynamic Time Warping emerging as a well-established solution for synchronizing motion sequences executed at different speeds.

Chapter 3

SYSTEM ARCHITECTURE

This chapter presents the high-level architecture of the AR-Based Kabaddi Ghost Trainer. The design philosophy, pipeline structure, data flow, and the separation of concerns that govern the modular organisation of the system are described.

3.1 ARCHITECTURAL OVERVIEW

3.1.1 SYSTEM DESIGN PHILOSOPHY

The system follows a modular, pipeline-driven processing model where each stage operates as an independent module with well-defined inputs and outputs, connected through a sequential data flow. The architecture is built on four core principles.

The first principle is "pose-first abstraction," where all evaluation is performed on extracted skeletal keypoints, thereby eliminating dependencies on lighting, background, or subject appearance. The second principle is "deterministic processing," meaning every metric computation is deterministic and identical inputs always produce identical outputs. The third principle is the "separation of evaluation and coaching," which ensures that numerical pose evaluation remains strictly separated from human-readable feedback generation. The fourth principle is "offline-first design," where the system defaults to offline batch processing and real-time modes are explicitly opt-in through configuration flags.

3.1.2 HIGH-LEVEL SYSTEM PIPELINE

The end-to-end pipeline comprises eight sequential stages. The first stage is the "input layer," which handles video recording of expert and user Kabaddi movements. The second stage, "pose extraction" (Level-1), performs person detection using YOLOv8 and keypoint estimation using MediaPipe, followed by MP33-to-COCO-17 format conversion. The third stage, "pose preprocessing," applies missing joint interpolation, pelvis centering, torso-based scale normalisation, outlier suppression, and EMA smoothing. The fourth stage, "temporal alignment" (Level-2), uses Dynamic Time Warping on pelvis trajectories to synchronise user and expert sequences. The fifth stage, "error localization" (Level-3), computes frame-wise Euclidean errors to produce joint-level error aggregates and phase decomposition. The sixth stage, "similarity scoring" (Level-4), combines structural and temporal similarity scores into an overall 0–100 score. The seventh stage, "feedback generation," serializes error metrics into JSON, transforms them into LLM prompts, and sends them to a local LLM for coaching feedback. The eighth and final stage, "AR visualization," renders ghost and user skeletons as video overlays.

3.1.3 ARCHITECTURAL BOUNDARIES AND ASSUMPTIONS

The system operates on pre-recorded video, and live capture falls outside the default scope. Pose estimation relies on external pre-trained models, namely YOLOv8 and MediaPipe, and no model training is performed within the system. All evaluation operates on 2D skeletal keypoints, and 3D reconstruction is not performed. A single subject per video is assumed, while multi-person scenes employ a motion-based raider selection heuristic to identify the primary subject.

3.1.4 SEPARATION OF VISUAL, ANALYTICAL, AND FEEDBACK LAYERS

The architecture enforces a strict three-layer separation. The "visual layer" handles AR rendering of ghost and user skeletons, producing video output without performing any evaluation. The "analytical layer" encompasses temporal alignment, error localization, and similarity scoring, operating on numerical pose arrays and producing structured metric dictionaries. The "feedback layer" transforms structured metrics into LLM-ready context, constructs prompts, and generates coaching feedback without directly accessing raw pose data.

3.2 OVERALL TECHNICAL ARCHITECTURE

3.2.1 SYSTEM ARCHITECTURE DIAGRAM

Figure 3.1 presents the complete system architecture, showing all major modules and their data flow relationships.

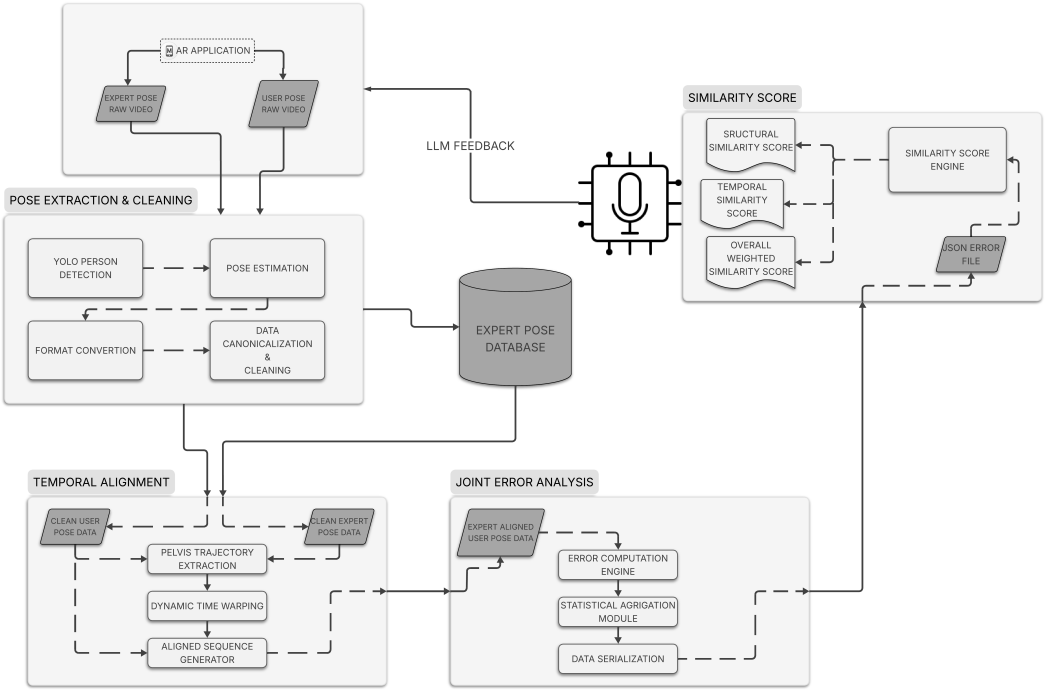

Figure 3.1: High-level system architecture of the AR-Based Kabaddi Ghost Trainer

The system comprises several principal components. The "Pipeline Runner" (`run_pipeline.py`) serves as the central orchestrator managing execution flow and output serialization. The "Pipeline Configuration" (`pipeline_config.py`) provides a centralized dataclass with configurable parameters. The "Pose Extraction Module" (`pose_extract_cli.py`) handles video ingestion, detection, estimation, format conversion, and cleaning. The "Temporal Alignment" module (`temporal_alignment.py`) performs pelvis-based DTW alignment. The "Error Localization" module (`error_localization.py`) computes frame-wise joint errors. The "Pose Validation Metrics" module (`pose_validation_metrics.py`) calculates structural and temporal scores. The "LLM Feedback System" (`llm_feedback/`) comprises the Context Engine, Prompt Builder, and LLM Client. Finally, the "AR Renderer" (`ar_pose_renderer.py`) generates skeletal visualizations.

3.2.2 DATA FLOW BETWEEN MODULES

Data flows in a strictly forward-propagating pattern with no cyclic dependencies. Video frames pass through person detection and pose estimation to produce raw poses of shape $(T, 33, 2)$. These are converted to COCO-17 format with shape $(T, 17, 2)$ and then processed through Level-1 cleaning. Expert and user cleaned poses are aligned through DTW to produce alignment indices. The aligned poses pass through error localization to produce error metrics, and then through similarity scoring to produce scores. Finally, scores flow through the Context Engine and Prompt Builder to the LLM, which generates feedback text. All pose data is communicated as NumPy arrays of shape $(T, 17, 2)$, and evaluation results are communicated as Python dictionaries.

3.2.3 OFFLINE VS ONLINE PROCESSING

The default execution mode is offline batch processing. Configuration flags control optional features: `enable_viz` controls AR visualization generation (enabled by default), `enable_tts` controls text-to-speech audio from feedback (disabled by default), and `verbose` controls logging verbosity.

3.3 SYSTEM MODULARITY AND EXTENSIBILITY

3.3.1 PLUG-AND-PLAY POSE ESTIMATOR

The pose extraction module accepts output from any pose estimation model, provided the output maps to COCO-17 format. The current implementation uses MediaPipe with a 33-to-17 mapping via `mp33_to_coco17.py`, but alternative models such as MoveNet or OpenPose can be integrated by implementing a corresponding format conversion module.

3.3.2 REPLACEABLE ALIGNMENT STRATEGY

The temporal alignment module exposes a well-defined interface: it accepts two pose sequences and returns aligned frame indices. Alternative strategies such as Canonical Time Warping, Soft-DTW, or fixed-rate interpolation can replace DTW by conforming to this interface.

3.3.3 LLM MODEL INTERCHANGEABILITY

The LLM model name, endpoint URL, and generation parameters are externalized in configuration. Switching between models such as Llama 3, Mistral, Gemma, or Qwen requires only a configuration change, as the prompt builder and context engine are model-agnostic.

3.4 COMPUTATIONAL CONSIDERATIONS

3.4.1 TIME COMPLEXITY

The pose extraction stage has a time complexity of $O(T \times F)$ and represents the most expensive stage, involving YOLOv8 and MediaPipe inference per frame. DTW alignment requires $O(T_{\text{user}} \times T_{\text{ghost}})$ for distance and cost matrix computation. Error localization and similarity scoring both operate at $O(T \times J)$ using vectorised Euclidean distance, where $J = 17$. LLM inference time is variable and dominated by autoregressive decoding time.

3.4.2 SPACE COMPLEXITY

The two pose arrays of shape $(T, 17, 2)$ contribute $O(T)$ space. The DTW cost and distance matrices of shape $(T_{\text{user}}, T_{\text{ghost}})$ require $O(T_{\text{user}} \times T_{\text{ghost}})$ space. The error matrix of shape $(T, 17)$ contributes $O(T)$ space. For typical video lengths of 100 to 300 frames, total memory usage remains well within standard hardware capacity.

3.5 DESIGN LIMITATIONS

The system has several inherent design limitations. First, all evaluation is restricted to 2D pose data, meaning depth information is unavailable and movements with significant depth variation may not be accurately captured. Second, pose quality depends on camera placement, resolution, and viewing angle, and a roughly frontal or lateral view with the full body visible is assumed. Third, the raider selection heuristic selects the most dynamically active person, and scenarios with multiple equally active subjects may result in incorrect selection. Fourth, the system evaluates structural and temporal similarity of body postures only, and tactical decision-making, reaction time, and contact force fall outside the evaluation scope.

Chapter 4

MODULE DESCRIPTION

This chapter provides a detailed description of each processing module in the system. For every module, the inputs, outputs, internal processing logic, and relevant implementation details are presented.

4.1 INPUT LAYER: DATA ACQUISITION MODULE

4.1.1 EXPERT AND USER CAPTURE PIPELINES

Both expert and user Kabaddi movements are recorded using a standard video camera or mobile device. The capture pipeline follows a two-stage approach applicable to both types of recordings. In the first stage, person detection is performed using YOLOv8 (nano variant, `yolov8n.pt`), which detects and tracks persons using the ByteTrack algorithm and assigns persistent IDs across frames. In the second stage, a raider selection heuristic identifies the individual with the highest cumulative motion magnitude as the primary raider in multi-person scenarios.

Figure 4.1 illustrates the person detection pipeline.

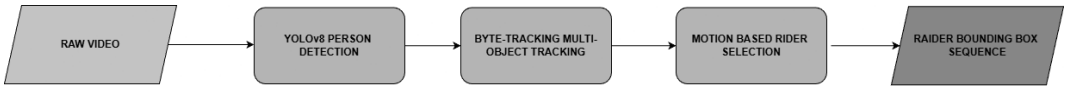

Figure 4.1: Person detection and raider selection pipeline

Both expert and user videos are processed through identical pipelines, ensuring no systematic bias in the comparison stage.

4.1.2 INPUT CONSTRAINTS

The input must be a valid video file readable by OpenCV, and at least one person must be visible in the majority of frames. A stationary or near-stationary camera is assumed, and the system expects a single primary subject (the raider) per video.

4.2 POSE EXTRACTION AND PREPROCESSING MODULE

4.2.1 HUMAN DETECTION AND POSE ESTIMATION

The selected person's bounding box is cropped with 40-pixel padding and passed to MediaPipe Pose with complexity level 1, detection confidence of 0.5, and tracking confidence of 0.7. MediaPipe outputs 33 body landmarks with normalised (x, y) coordinates. Figure 4.2 shows this flow.

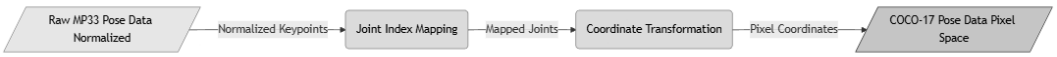

Figure 4.2: Pose estimation flow: bounding box cropping to raw keypoint extraction

4.2.2 KEYPOINT REPRESENTATION FORMAT

The system uses the COCO-17 keypoint format as its canonical representation. The 17 keypoints comprise: Nose (0), Left/Right Eye (1–2), Left/Right Ear (3–4), Left/Right Shoulder (5–6), Left/Right Elbow (7–8), Left/Right Wrist (9–10), Left/Right Hip (11–12), Left/Right Knee (13–14), and Left/Right Ankle (15–16). Each keypoint is represented by (x, y) coordinates, producing a per-sequence tensor of shape $(T, 17, 2)$.

4.2.3 POSE CLEANING PIPELINE

The Level-1 cleaning pipeline (`level1_cleaning.py`) applies deterministic transformations in strict sequential order. The first half of the pipeline is shown in Figure 4.3 and the second half in Figure 4.4.

Figure 4.3: Level-1 cleaning: temporal interpolation and pelvis centering

The cleaning pipeline comprises six sequential stages. First, “invalid joint detection” flags joints with non-finite values or zero-sum coordinates. Second, “temporal interpolation” linearly interpolates missing joints between valid

Figure 4.4: Level-1 cleaning: scale normalisation, outlier suppression, and EMA smoothing

observations, requiring a minimum of two anchors. Third, "pelvis centering" translates the pelvis midpoint (the average of left and right hip coordinates) to the origin at each frame. Fourth, "scale normalisation" divides all coordinates by the torso length (shoulder midpoint to hip midpoint distance), with a safety factor of $\varepsilon = 10^{-6}$. Fifth, "outlier suppression" replaces frames with velocity magnitudes exceeding 3σ with the previous frame's data. Sixth, "EMA smoothing" applies an exponential moving average with $\alpha = 0.75$ to attenuate high-frequency noise.

The output is a canonical pose array of shape $(T, 17, 2)$ that is pelvis-centred and torso-normalised, serving as the standard input for all downstream modules.

4.3 EXPERT POSE DATABASE

Expert reference motions, referred to as "ghosts," are stored as NumPy `.npy` files with shape $(T, 17, 2)$ using 64-bit floating-point precision. Each file corresponds to a specific Kabaddi technique. For a typical 150-frame sequence,

storage is approximately 40 KB. The pipeline runner accepts expert pose paths as command-line arguments.

4.4 TEMPORAL ALIGNMENT MODULE

4.4.1 PELVIS TRAJECTORY AS TEMPORAL ANCHOR

Human movements exhibit natural speed variations even when performing the same technique. Direct frame-to-frame comparison without alignment would penalise speed differences as positional errors. The pelvis trajectory is used as the primary temporal anchor because it represents the body's approximate centre of mass and provides a stable, low-noise signal. Figure 4.5 shows the trajectory extraction process.

Figure 4.5: Pelvis trajectory extraction from cleaned pose sequences

4.4.2 DYNAMIC TIME WARPING ARCHITECTURE

The DTW algorithm finds the optimal warping path that minimises total alignment cost while enforcing boundary, monotonicity, and continuity constraints. The pairwise distance matrix between pelvis positions is computed as:

$$D[i,j] = \sqrt{(x_i^{\text{user}} - x_j^{\text{ghost}})^2 + (y_i^{\text{user}} - y_j^{\text{ghost}})^2}$$

The accumulated cost matrix is then computed using dynamic programming:

$$C[i,j] = D[i,j] + \min\big(C[i-1,j],\ C[i,j-1],\ C[i-1,j-1]\big)$$

The optimal path is recovered through backtracking from $(T_{\text{user}} - 1, T_{\text{ghost}} - 1)$ to $(0,0)$. Figure 4.6 illustrates this process.

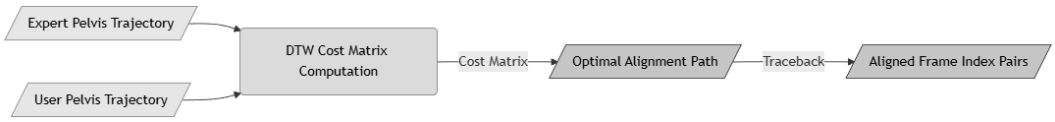

Figure 4.6: DTW cost matrix computation and optimal alignment path recovery

4.4.3 ALIGNED SEQUENCE GENERATION

The DTW warping path produces index pairs establishing temporal correspondence between user and expert frames. These indices are used to construct aligned sequences of equal effective length for direct comparison. DTW naturally handles speed variability through many-to-one and one-to-many frame correspondences. Figure 4.7 shows the alignment output.

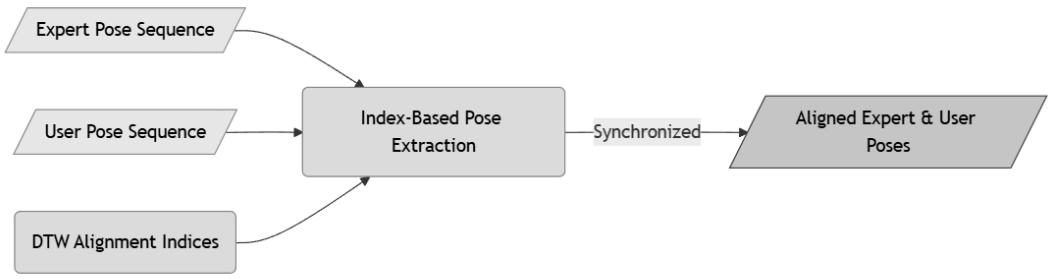

Figure 4.7: Index-based pose extraction producing aligned expert and user sequences

4.5 JOINT-LEVEL ERROR COMPUTATION MODULE

4.5.1 ERROR MATRIX CONSTRUCTION

The error localization module computes frame-wise joint errors between aligned sequences. For each frame t and joint j, the error is computed as:

$$e_{t,j} = \left\| \mathbf{p}_{t,j}^{\text{user}} - \mathbf{p}_{t,j}^{\text{expert}} \right\|_2$$

This produces an error matrix $E \in \mathbb{R}^{T \times 17}$, computed using vectorised NumPy operations. Figure 4.8 illustrates the computation.

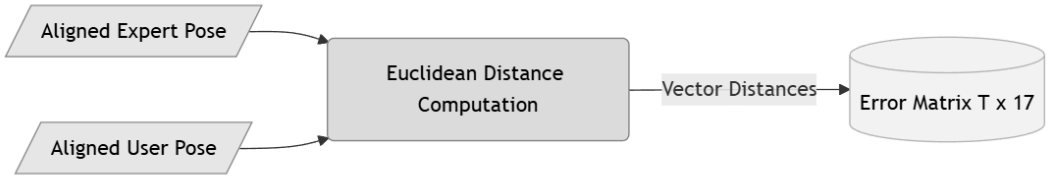

Figure 4.8: Euclidean distance computation between aligned pose pairs

4.5.2 STATISTICAL AGGREGATION

For each joint, three statistics are computed across all frames: mean error, maximum error, and standard deviation. Figure 4.9 shows the aggregation into joint, frame, and phase statistics.

4.5.3 JOINT IMPORTANCE WEIGHTING

Kabaddi-specific joint weights prioritise biomechanically significant joints such as shoulders, elbows, wrists, hips, knees, and ankles over facial landmarks such as the nose, eyes, and ears. Weighted spatial distances ensure that errors in technique-critical joints contribute proportionally more to the overall score.

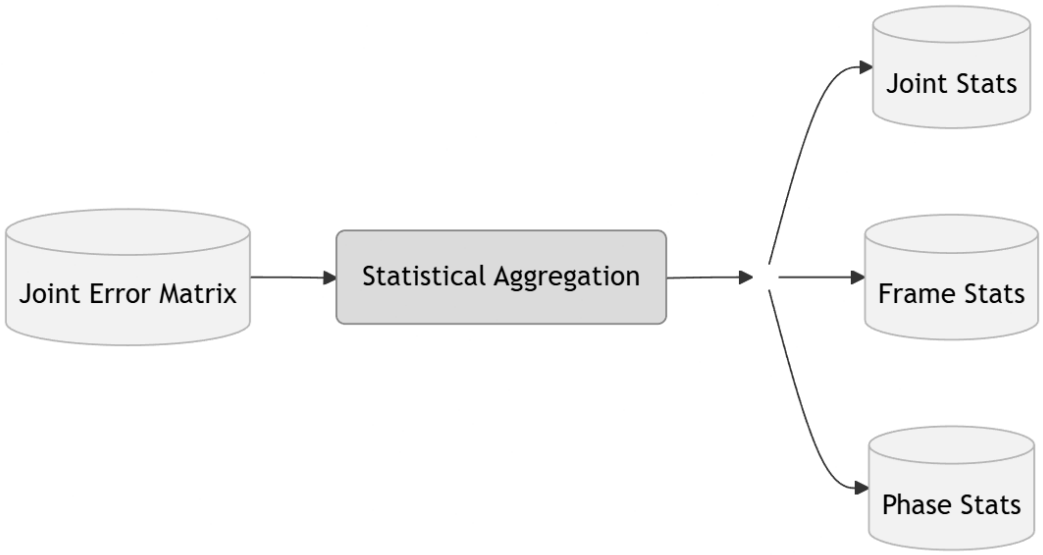

Figure 4.9: Statistical aggregation of the error matrix into joint, frame, and phase statistics

4.6 PHASE-WISE TEMPORAL ERROR ANALYSIS

4.6.1 MOVEMENT PHASE SEGMENTATION

The aligned sequence is divided into three equal-duration temporal phases. The "early phase" covers frames $[0, T/3)$ and corresponds to movement initiation and preparation. The "mid phase" covers frames $[T/3, 2T/3)$ and corresponds to primary execution. The "late phase" covers frames $[2T/3, T)$ and corresponds to completion and recovery. Per-phase, per-joint mean errors are computed to identify which joints exhibit the highest deviations during each phase.

4.6.2 TEMPORAL TREND DETECTION

The Context Engine classifies temporal trends by comparing early-phase and late-phase errors. A trend is classified as "improving" when the late error is less than the early error beyond a threshold of $\delta = 0.1$, as "degrading" when the late error

exceeds the early error beyond this threshold, or as "stable" when the difference falls within δ.

4.7 SIMILARITY SCORE ENGINE

4.7.1 STRUCTURAL SIMILARITY SCORE

The structural similarity score combines two components. The "weighted spatial distance" component, contributing 70% of the structural score, computes the Kabaddi-weighted Euclidean distance across aligned frames and converts it to a 0–100 scale via exponential decay. The "angular consistency" component, contributing 30%, computes the mean angular deviation across key joint triplets such as shoulder–elbow–wrist. The structural score is expressed as:

$$S_{\text{structural}} = 0.7 \times S_{\text{spatial}} + 0.3 \times S_{\text{angular}}$$

4.7.2 TEMPORAL SIMILARITY SCORE

The temporal similarity score is computed using the DTW distance between flattened pose sequences, normalised and scaled to 0–100:

$$S_{\text{temporal}} = \max\left(0,\ 100 \times \left(1 - \frac{D_{\text{DTW}}}{D_{\max}}\right)\right)$$

4.7.3 OVERALL SIMILARITY SCORE

The overall similarity score is a weighted combination of the structural and temporal components:

$$S_{\text{overall}} = 0.7 \times S_{\text{structural}} + 0.3 \times S_{\text{temporal}}$$

Figure 4.10 shows the scoring pipeline.

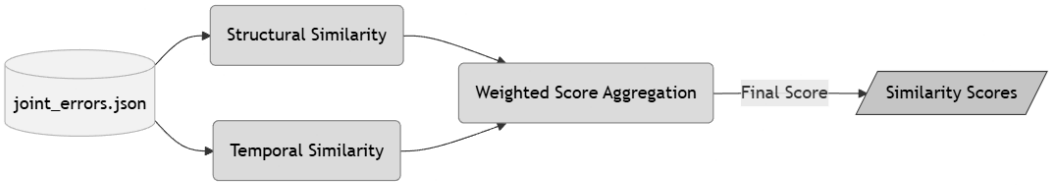

Figure 4.10: Similarity scoring: structural and temporal components combined into final score

4.7.4 SCORE INTERPRETATION

Scores are mapped to qualitative categories: “Excellent” for scores of 90 or above, “Good” for scores between 75 and 89, “Fair” for scores between 60 and 74, and “Needs Improvement” for scores below 60.

4.8 ERROR SERIALIZATION AND STRUCTURED OUTPUT

All evaluation results are serialized into structured JSON for downstream consumption. Figure 4.11 shows the data packaging process.

4.8.1 SEVERITY CLASSIFICATION

Joints are classified by mean error magnitude into three tiers: “major” for mean errors exceeding 0.7, “moderate” for mean errors between 0.3 and 0.7, and

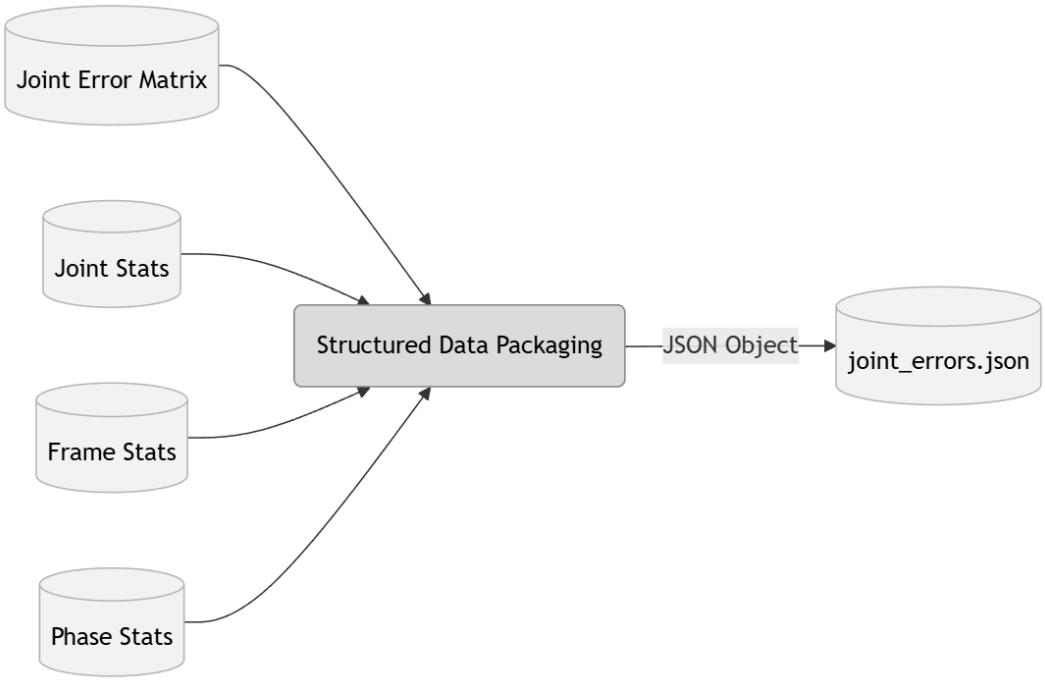

Figure 4.11: Structured data packaging: error matrices and statistics serialized to JSON

"minor" for mean errors at or below 0.3. Temporal phases are similarly tagged as "Excellent" (mean error ≤ 0.3), "Good" (0.3–0.5), "Fair" (0.5–0.8), or "Poor" (above 0.8), with the top three dominant joints identified per phase.

4.9 LLM-BASED FEEDBACK MODULE

4.9.1 CONTEXT BUILDER

The Context Engine (`context_engine.py`) transforms raw pipeline output into canonical JSON through score classification, joint deviation tiering, phase quality assessment, and temporal trend detection. The output is a self-contained context object that serves as the sole input for prompt construction.

4.9.2 PROMPT COMPOSER

The Prompt Builder (`prompt_builder.py`) constructs prompts using a template-based approach. A system prompt defines the role of the LLM, and an instruction template is populated with context fields. Field consumption rules enforce that all major deviations are included, at most three moderate deviations are reported, and minor deviations are excluded.

4.9.3 LLM CLIENT

The LLM Client (`llm_client.py`) communicates with a locally hosted LLM via the Ollama HTTP API (`http://localhost:11434/api/generate`). The model name, temperature (default: 0.3), token limits, and timeouts are configurable. Both streaming and non-streaming modes are supported. This vendor-agnostic design allows model switching without code changes.

4.9.4 FEEDBACK PIPELINE SUMMARY

The feedback pipeline operates in three stages. In the first stage, raw scores are processed by the Context Engine to produce canonical context JSON. In the second stage, the Prompt Builder consumes this context to produce system and instruction prompts. In the third stage, the LLM Client sends these prompts to the locally hosted LLM and returns the generated coaching feedback. The evaluation and coaching pipelines share no internal state, ensuring that feedback strategy changes never affect numerical evaluation.

4.9.5 TECHNIQUE-AWARE PROMPTING

The instruction prompt includes the specific Kabaddi technique name (e.g., "Bonus Step") passed from the frontend via the session context. This allows the

LLM to generate technique-specific coaching advice rather than generic movement feedback. The technique name is injected into the instruction template as a dedicated field and appears at the top of the prompt context, ensuring the LLM anchors its response to the correct movement type.

4.9.6 RAW LLM BASELINE FOR COMPARISON

To demonstrate the effectiveness of the prompt engineering pipeline, the system includes a raw LLM inference mode. In this mode, the same LLM model receives a generic prompt — "Give me feedback on my kabaddi raid technique" — without any performance context, system prompt constraints, or structured data. This produces a baseline response that can be directly compared against the prompt-engineered output, allowing empirical evaluation of how context injection and prompt design improve feedback quality.

4.9.7 AUTOMATED FEEDBACK QUALITY METRICS

The system implements an automated evaluation framework (`feedback_metrics.py`) that scores LLM-generated feedback across five quality dimensions. Each metric is scored on a scale of 1 to 5 based on deterministic text analysis of the generated feedback against the source context data. Table 4.1 summarises the five metrics.

Table 4.1: Automated feedback quality metrics

Metric	**Description**	**Scoring**
Groundedness	References actual joints, phases, and assessments from context	Higher = Better (1–5)
Hallucination	Invents joints or facts not present in the data	Lower = Better (1–5)
Specificity	Uses specific joint names vs. generic advice terms	Higher = Better (1–5)
Relevance	Tone matches the actual performance score level	Higher = Better (1–5)
Technique Awareness	References the specific technique name and domain terms	Higher = Better (1–5)

GROUNDEDNESS

Groundedness measures whether the feedback references actual data from the context. Let J_{ctx} be the set of joints flagged as major or moderate deviations in the context, J_{ref} be the subset of J_{ctx} that are mentioned in the feedback text, and define two boolean indicators: $a = 1$ if the overall assessment label (e.g., "Fair") appears in the feedback, and $p = 1$ if the temporal trend pattern (e.g., "improving") appears in the feedback. The groundedness ratio is computed as:

$$r_G = \frac{|J_{\text{ref}}| + a + p}{|J_{\text{ctx}}| + 2}$$

The score is then mapped to a 1–5 scale: $G = \max(1, \min(5, \lfloor r_G \times 5 \rceil))$.

HALLUCINATION DETECTION

Hallucination detection identifies claims in the feedback that are not supported by the context data. The metric counts two types of hallucinations. First, *hallucinated joints*: joint names from the COCO-17 vocabulary that appear in the feedback but are not flagged in any severity tier (major, moderate, or minor) of the context. Let $J_{\text{mentioned}}$ be the set of all COCO-17 joint names detected in the feedback, and $J_{\text{all_ctx}}$ be the set of all joints referenced in the context across all severity tiers. The hallucinated joint set is $J_{\text{halluc}} = J_{\text{mentioned}} \setminus J_{\text{all_ctx}}$. Second, *fabricated numeric claims*: instances of numeric values followed by units such as percentages, degrees, or angles, detected using the regular expression pattern `\d+[\.d]* (%|degrees?|percent)'`. The total hallucination count is:

$$h = |J_{\text{halluc}}| + n_{\text{numeric}}$$

The score is assigned using a step function: $H = 1$ if $h = 0$, $H = 2$ if $h = 1$, $H = 3$ if $h \leq 3$, $H = 4$ if $h \leq 5$, and $H = 5$ otherwise. A lower H indicates better

performance (fewer hallucinations).

SPECIFICITY

Specificity quantifies how precise the feedback is by comparing the use of specific anatomical references against vague generic terms. Let n_s be the count of distinct COCO-17 joint names (e.g., "left wrist", "right ankle") found in the feedback, n_g be the count of generic terms matched from a predefined list of 11 vague phrases (e.g., "your body", "some areas", "various joints"), and n_p be the count of phase-specific terms ("early", "mid", "late"). The specificity index is:

$$s = n_s + n_p - n_g$$

The score is mapped as: $S = 5$ if $s \geq 5$, $S = 4$ if $s \geq 3$, $S = 3$ if $s \geq 1$, $S = 2$ if $s \geq 0$, and $S = 1$ otherwise.

RELEVANCE

Relevance verifies whether the tone of the feedback matches the actual performance level. The context provides an overall assessment label — "Excellent", "Good", "Fair", or "Needs Improvement" — based on the similarity scores. Each label is associated with a predefined set of expected tone words. For example, "Excellent" maps to words like "outstanding", "perfect", and "superb", while "Needs Improvement" maps to "poor", "weak", and "struggling".

The metric performs two checks. First, it counts *matching tone words m* — expected tone words for the actual assessment that appear in the feedback. Second, it counts *mismatched tone words d* — tone words from the opposite end of the assessment spectrum (e.g., "excellent" appearing in "Needs Improvement" feedback, or "poor" appearing in "Excellent" feedback).

The score is assigned as: $R = 5$ if $m \geq 2$ and $d = 0$; $R = 4$ if $m \geq 1$ and $d = 0$; $R = 3$ if $m \geq 1$ and $d \leq 1$; $R = 2$ if $d \leq 2$; and $R = 1$ otherwise.

TECHNIQUE AWARENESS

Technique Awareness checks whether the feedback references the specific Kabaddi technique being evaluated. Let t_{full} be a boolean indicating whether the full technique name (e.g., "Bonus Step") appears in the feedback. Let W_t be the set of individual words in the technique name (excluding words with ≤ 2 characters), and W_{found} be the subset present in the feedback. Additionally, let K be the set of Kabaddi-specific terms — "raid", "raider", "kabaddi", "touch", "bonus", "cant" — found in the feedback.

The score is assigned as: $T = 5$ if $t_{\text{full}} = 1$ and $|K| \geq 1$; $T = 4$ if $t_{\text{full}} = 1$; $T = 3$ if $|W_{\text{found}}| \geq 1$ and $|K| \geq 1$; $T = 2$ if $|K| \geq 1$; and $T = 1$ otherwise.

COMPOSITE QUALITY SCORE

The five individual metrics are combined into a single composite quality percentage. Since the hallucination metric uses an inverted scale (lower is better), it is transformed before summation. The composite score is:

$$Q = \frac{G + (6 - H) + S + R + T}{25} \times 100$$

where G = Groundedness, H = Hallucination (inverted via $6 - H$ so that low hallucination contributes high points), S = Specificity, R = Relevance, and T = Technique Awareness. Each component contributes equally with a maximum of 5 points, yielding a maximum possible composite score of 25 (i.e., 100%).

The comparison endpoint (`/llm_feedback/compare/`) generates both the prompt-engineered and raw responses in a single request, computes all five metrics

on each, and returns the results with per-metric winners for side-by-side display in the frontend.

4.10 AUGMENTED REALITY VISUALIZATION MODULE

The AR Pose Renderer (`ar_pose_renderer.py`) provides OpenCV-based skeletal visualization in three rendering modes. In the "ghost rendering" mode, the expert skeleton is drawn in green on a black canvas. In the "user rendering" mode, the user skeleton is drawn in blue. In "overlay mode," both skeletons are rendered simultaneously for visual deviation comparison.

Video output uses the H.264 codec (`mp4v`). When sequences differ in length, temporal interpolation synchronises them before overlay. A frame export mode is also available for debugging purposes.

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